26 Each student at a college is going to be given a notebook.

The colour of each notebook will be red or blue or green or orange.

Bill takes a sample of 50 of the students at the college.
He asks each of these students what colour of notebook they want.

The table gives Bill's results.

Notebook colour	red	blue	green	orange
Number of students	18	16	10	6

There are 3000 students at the college.

(a) Work out how many red notebooks Bill should buy.

1 mark
Scale sample to3000

	Red	All
	18	50
×60		×60
	1080	3000

3000÷50=60 samples.

1 mark
1080

1080 red notebooks.

(2)

(b) Write down one assumption that you made in answering part (a).
Explain how your answer would be affected if this assumption were not true.

Assume the sample represents the whole college.

1 mark
Assumption and effect

If it does not, the proportion wanting red changes,

so he may need more or fewer than1080 red notebooks.

(1)

(Total for Question 26 is 3 marks)